

# Use case: AI-assisted returns at NovaRetail

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NovaRetail is a medium-sized online and physical retailer selling electronics, home products, and clothing.

The company processes approximately **12,000 customer-service requests per month**. Around 40% concern returns, refunds, damaged products, delivery problems, or warranty questions.

Customer-service agents currently need to search across:

- the customer relationship management system;
- order and delivery systems;
- product and warranty databases;
- return policies;
- previous emails;
- spreadsheets maintained by individual departments.

The process is slow and inconsistent:

- average response time: **18 hours**;
- first-contact resolution: **62%**;
- approximately **28% of cases are escalated**;
- refund decisions are sometimes inconsistent;
- agents spend considerable time copying information between systems;
- customers complain that different agents provide different answers.

The Chief Operating Officer proposes:

“Let us implement an AI customer-service system that automatically answers customers and approves simple refunds.”

The customer-service manager is interested but concerned about incorrect answers, fraudulent refund requests, employee resistance, and loss of customer trust.

The company has a limited budget and wants to run a **six-week pilot** before making a larger investment.

## Available information

NovaRetail has:

- three years of customer-service conversations;
- order, payment, and delivery records;
- written return and warranty policies;
- product information;
- customer satisfaction scores;
- records of refunds and escalations.

However:

- some policies are outdated;
- different departments use different definitions;
- historical responses contain mistakes;
- customer-service data includes personal information;
- refund decisions are not always documented;
- the systems are not fully integrated.

The company is considering several possible interventions:

1. an AI assistant that searches policies and drafts responses for agents;
2. an AI chatbot that answers customers directly;
3. an AI system that recommends whether a refund should be approved;
4. a system that automatically approves low-value refunds;
5. a simpler search and workflow tool without generative AI.

The team must decide whether AI should be used, where it should be used, and under what conditions.

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## Student task: complete the AI Implementation Canvas

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Your team must design a responsible AI intervention for NovaRetail. You may recommend an AI solution, a simpler technology solution, process redesign, a limited pilot, or no implementation.

Complete the canvas by answering the following questions.

### 1. Strategic objective

What important business or organizational outcome should NovaRetail improve?

Examples include:

- reducing response time;
- improving consistency;
- reducing unnecessary escalations;
- improving customer satisfaction;
- reducing refund losses;
- improving employee productivity.

## **2. Business problem**

What is the precise problem?

Who experiences it, and what is the current cost of the problem?

Distinguish between symptoms and root causes.

## **3. Current process**

Map the existing returns or customer-service process.

Show:

- customer request;
- information required;
- decisions made;
- systems consulted;
- human roles;
- exceptions and escalations;
- final action.

## **4. Users and stakeholders**

Identify the main stakeholders:

- customers;
- customer-service agents;
- customer-service managers;
- finance;
- logistics;
- legal and compliance;
- IT and security;
- senior management.

What does each stakeholder want, and what might each stakeholder fear?

## **5. Role of AI**

Define precisely what the AI system would do.

For example, would it:

- retrieve relevant policies;
- summarize a customer case;
- classify the request;
- draft a response;
- recommend an action;
- predict fraud risk;
- approve a refund;
- communicate directly with the customer?

What should the AI explicitly not do?

## **6. Human–AI interaction**

Specify:

- what AI does;
- what the employee does;
- when human approval is required;
- when the case must be escalated;
- how employees correct the system;
- who remains accountable for the final decision.

## **7. Data and context**

Identify the data required.

For each source, consider:

- accuracy;
- completeness;
- freshness;
- access permissions;
- privacy;
- conflicting information;
- ownership;
- maintenance responsibility.

What information would the AI need in order to provide a reliable answer?

## **8. Workflow redesign**

Show the proposed future workflow.

Explain how the process would change for:

- ordinary cases;
- ambiguous cases;
- high-value refunds;
- suspected fraud;
- vulnerable customers;
- policy exceptions;
- system failure.

## **9. Evaluation plan**

Define how the pilot would be tested.

Include:

- a baseline;
- representative test cases;
- difficult and unusual cases;
- error categories;
- human evaluation;
- customer impact;
- acceptance criteria.

Possible metrics include:

- response time;
- first-contact resolution;
- escalation rate;
- factual accuracy;
- policy compliance;
- refund error rate;
- customer satisfaction;
- agent correction rate.

## **10. Risks and safeguards**

Identify at least five risks.

Consider:

- hallucinated policies;
- incorrect refunds;
- discriminatory treatment;
- privacy violations;
- prompt injection;
- manipulation by customers;
- over-reliance by employees;
- unclear accountability;
- outdated information;
- reputational damage.

For each risk, propose prevention, detection, and recovery measures.

## **11. Business value and economics**

Estimate:

- expected benefits;
- implementation costs;
- model and infrastructure costs;
- integration costs;
- employee training costs;
- monitoring and maintenance costs;
- cost of errors;
- expected return on investment.

Compare the proposed AI solution with the no-AI alternative.

## **12. Pilot and implementation plan**

Design the smallest credible six-week pilot.

Specify:

- pilot users;
- pilot process;
- excluded cases;
- systems required;
- safeguards;
- training;
- monitoring;
- review points;

- decision at the end of the pilot.

## 13. Final recommendation

Choose one:

- implement a controlled pilot;
- redesign the process before piloting;
- build the data and knowledge foundation first;
- use a simpler non-AI solution;
- collect more evidence;
- postpone;
- do not implement AI.

Your recommendation must be justified by evidence, assumptions, risks, and measurable conditions.

## Final deliverable

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Submit:

- the completed AI Implementation Canvas;
- one current-state process map;
- one future-state workflow;
- a pilot evaluation plan;
- a risk-and-safeguard table;
- a final recommendation of approximately 500 words.

The central question is:

Should NovaRetail allow AI to answer customers and make refund decisions, or should it use AI in a narrower, more controlled role?